

FARM ANIMALS

Question

Classes and Objects CO1

Design a “farm animals” java application with the details of animals like cow, pig, horse.

Consider the following details like where they stay, what they eat, the sound they make by

using classes and objects.

Step wise execution

1. The program starts execution from the main() method.

2. A cow object is created using:

```
Animal("cow", "shed", "grass", "amba").
```

3. The constructor assigns:

```
name = cow
```

```
place = shed
```

```
food = grass
```

```
sound = amba
```

4. cow.showAnimaldetails() is called.

5. The cow details are displayed:

```
Animal: cow
```

```
place: shed
```

```
eats: grass
```

```
sound: AMBA
```

6. A pig object is created using:

```
Animal("pig", "mudd", "corn", "oink").
```

7. The constructor assigns:

```
name = pig
```

```
place = mudd
```

```
food = corn
```

```
sound = oink
```

8. pig.showAnimaldetails() is called.

9. The pig details are displayed:

```
Animal: pig
```

```
place: mudd
```

```
eats: corn
```

```
sound: OINK
```

10. A horse object is created using:

```
Animal("horse", "shed", "grass", "neighhh").
```

11. The constructor assigns:

```
name = horse
```

```
place = shed
```

```
food = grass
```

```
sound = neighhh
```

12. horse.showAnimaldetails() is called.

13. The horse details are displayed:

```
Animal: horse
```

place: shed

eats: grass

sound: NEIGHHH

14. After displaying all animal details, the program terminates.

Algorithm

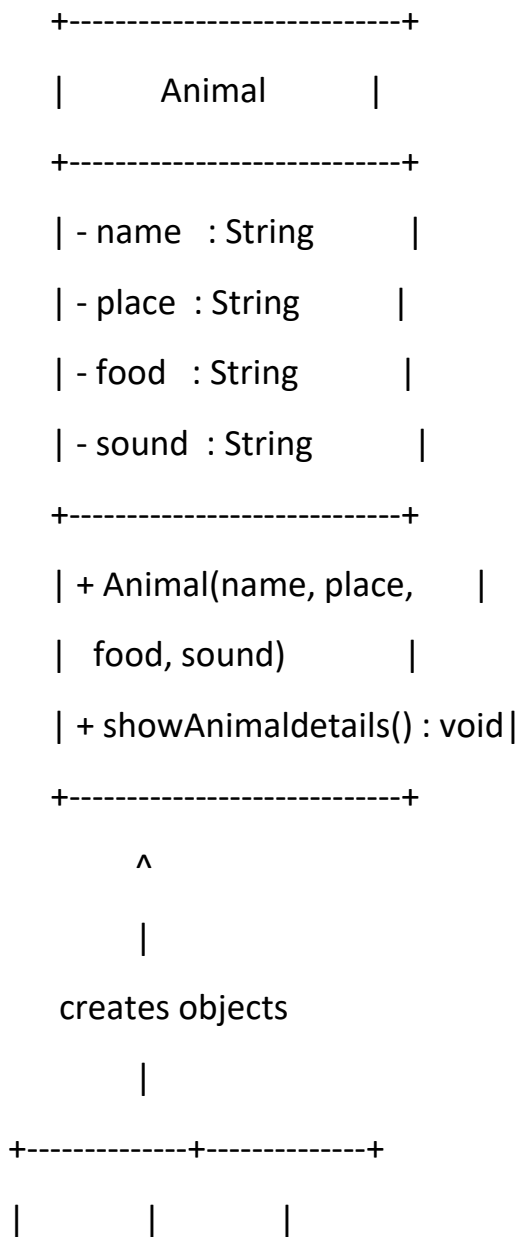
1. Start the program.
2. Define an Animal class with four variables:
name, place, food, and sound.
3. Create a constructor to initialize the animal details.
4. Define the showAnimaldetails() method to display the animal's name, place, food, and sound.
5. Start the main() method.
6. Create a cow object with the values:
cow, shed, grass, amba.
7. Call showAnimaldetails() to display the cow's details.
8. Create a pig object with the values:
pig, mudd, corn, oink.
9. Call showAnimaldetails() to display the pig's details.
10. Create a horse object with the values:

horse, shed, grass, neighhh.

11. Call showAnimaldetails() to display the horse's details.

12. Stop the program.

UML diagram



+-----+	+-----+	+-----+
cow	pig	horse
+-----+	+-----+	+-----+
shed	mudd	shed
grass	corn	grass
amba	oink	neighhh
+-----+	+-----+	+-----+

Code

```
package basics;

class Animal{
    String name;
    String place;
    String food;
    String sound;

    public Animal(String name, String place, String food,
String sound){
        this.name=name;
        this.place=place;
        this.food=food;
        this.sound=sound;
    }

    void showAnimaldetails(){
        System.out.println("Animal:" + name);
        System.out.println("place:" + place);
        System.out.println("eats:" +food);
        System.out.println("sound:" + sound.toUpperCase() + "\n");
    }
}
```

```

    }
}

public class FarmAnimals {
    public static void main(String[] args) {
        Animal cow=new Animal("cow","shed", "grass","amba" );
        cow.showAnimaldetails();
        Animal pig=new Animal("pig","mudd","corn","oink");
        pig.showAnimaldetails();
        Animal horse=new
Animal("horse","shed","grass","neighhh");
        horse.showAnimaldetails();
    }
}

```

Output screen shot

```

PS C:\Users\Admin\OneDrive\Documents\basicjava> c:: cd 'c:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
deDetailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.FarmAnimals'
Animal:cow
place:shed
eats:grass
sound:AMBA

Animal:pig
place:mudd
eats:corn
sound:OINK

Animal:horse
place:shed
eats:grass
sound:NEIGHHH

```